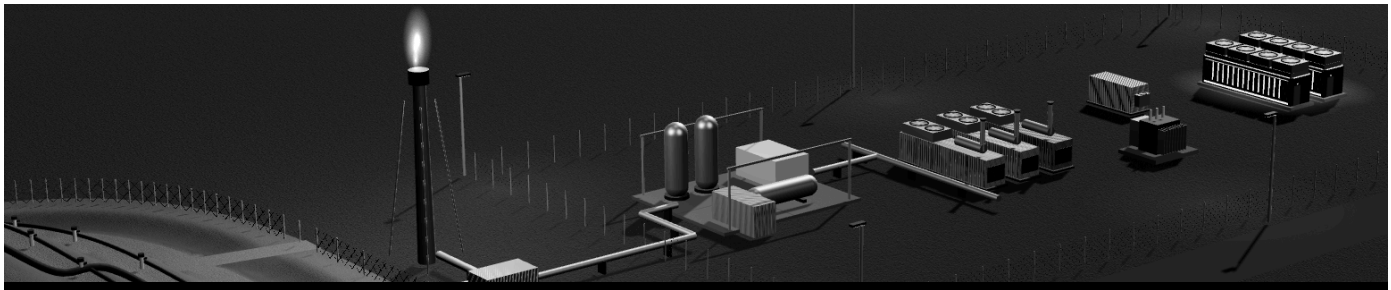
Butterfield Station Landfill, Mobile, Maricopa County, Arizona: a first sizing

What the gas the site now flares could power, how much land it would take, and what it could earn each year.

2.1 MW	2	7 ac	190k
POWER FROM THE GAS	COMPUTE MODULES	LAND NEEDED	MMBTU BOUGHT / YR

GAS SALE, PER YEAR	LAND LEASE, PER YEAR	TOTAL TO THE OWNER, PER YEAR
up to $571k	$7k to $15k	$7k to $586k



YOUR SITE IN OUR LIVE 3D MODEL — ACCRETIONCOMPUTE.COM/SITE-MODEL?SRC=LFG&FLOW=876&CH4=43

THE MATH

Landfill gas collected and flared (2022)	876 scfm
Methane content (2022)	43%
Energy in the gas	22.9 MMBtu per hour
Power at a heat rate of 11,000 Btu/kWh	about 2.1 MW
Compute (power less cooling and losses, /1.2)	about 1.7 MW
Compute modules (about 1 MW each)	2
Gas engines (running plus one spare)	3 at 1.6 MW
Land: 5 acres plus 1.2 acres per MW	about 7 acres
Gas bought per year (95% run time)	190k MMBtu
Gas payment, up to $3.00/MMBtu	up to $571k
Land lease at $1,000 to $2,000 per acre per year	$7k to $15k

Landfill gas: about 1,012 Btu per cubic foot of methane. Prices in US dollars; the price band is the one shown on our Site model page.

WHAT IT WOULD LOOK LIKE

- Gas engines on a small parcel near the gas, a battery to smooth the load, and factory-built compute modules leased to an AI company.
- No grid connection and no export: all of the power is used on site.
- We build and run the site. The owner puts in no capital.
- Multi-year gas and land agreements, matched to the tenant's lease, at a fixed price.

WORTH CHECKING

- Latest flow and methane data: intake of about 1.2 million tons a year points to more gas each year
- Whether the gas is free of other plans
- Space for a small fenced parcel near the flare
- Gas quality (siloxanes, H2S) for engines
- Maricopa County land use rules for the parcel

Source: EPA Landfill Methane Outreach Program (LMOP) database, September 2024 release (2022 data): about 1.26 million cubic feet a day (876 scfm) flared at 43% methane; no energy project on record. The site takes about 1.2 million tons a year, so gas should grow. This is a first read from public data, not an offer: no cost and no obligation. The real price and terms are set in the gas and land agreements after a site review.